

PHASE 04

NPG Annealings

QUICK OPERATOR GUIDE
Release 0.9.36 · Build 2026.08.11-r15

PURPOSE

Run the final oven annealing recipe while ramping the Phase 03 Keysight handoff current down independently and safely.

WORKFLOW AT A GLANCE

1 · START Accept Phase 03 handoff	2 · INITIAL 200 C for 5 min	3 · STAGE 1 350 C for 15 min	4 · STAGE 2 600 C for 40 min	5 · COOLDOWN PID 0 C for 10 min, save and finish
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BEFORE STARTING

- ✓ Verify the shutter is closed after Phase 03.
- ✓ Confirm the expected Keysight handoff: about 0.640 A with OUTPUT ON.
- ✓ Check oven PID, CK-1 temperature and recipe targets before starting.

CONTROLS AND END STATES

Normal completion	Keysight reaches 0 A and OUTPUT OFF; oven stays at PID setpoint 0 C.
Abort / Safe Stop	Requests oven PID 0 C and forces Keysight OUTPUT OFF; closes Phase 04 only.

KEY FACTS

Keysight ramp-down	Runs in parallel with the oven recipe.
Final PID command	0 C - there is no final 30 C command.
Final hold	10 min at setpoint 0 C
Automatic finish	Save outputs, then close Phase 04.

RUN DATA

Saved under:
Data Samples/NPG Annealing Data/

SAFETY - VERIFY THE REAL HARDWARE

Software completion does not mean the chamber is physically cool. Confirm safe real temperatures and the electrical state before venting or handling the sample.